

Algebra II - Quiz 1 - 25 Marks

February 9, 2024: Time: One hour

1. Let H and K be normal subgroups such that $H \cap K = \{1_G\}$. Show that $hk = kh$ for all $h \in H$ and $k \in K$. (5 marks)
2. What are all possible non-trivial homomorphisms from S_4 to $Z_2 \times Z_2$. (5 marks)
3. Let G be a group. An automorphism is an isomorphism $\phi : G \longrightarrow G$. Let $\text{Aut}(G) : \{\phi : G \longrightarrow G \mid \phi \text{ is an isomorphism}\}$.
 - (a) Show that $\text{Aut}(G)$ is a group. (5 marks)
 - (b) Determine $\text{Aut}(\mathbb{Z}_n)$. (5 marks)
4. Let $a \in \mathbb{C}$. Determine whether the two elements $\begin{pmatrix} a+1 & 0 \\ 0 & a \end{pmatrix}$ and $\begin{pmatrix} a-1 & a-1 \\ -a & 3a \end{pmatrix}$ are conjugate to each other as elements of $GL_2(\mathbb{C})$ (5 marks)